
Reactive vs Septic Arthritis

RHEUMATOLOGY / INFECTIOUS DISEASE

Two joints, same presentation, opposite mechanisms

A swollen, painful joint shows up the same way at the bedside whether it's reactive arthritis or disseminated gonococcal infection: migratory polyarthralgia, tenosynovitis, systemic symptoms. But inside the joint, the two conditions are mechanistically opposite. Reactive arthritis is sterile, a T-cell attack that outlives the infection that trained it. Disseminated gonococcal infection is a live bacterial invasion that seeded the joint through the bloodstream.



Baseline joint capsule

Molecular mimicry: the case of mistaken identity

HLA-B27 presents a fragment of the original organism to a T-cell, training that T-cell to attack it. The problem: a joint self-antigen looks close enough to that fragment that the same T-cell attacks the joint's own tissue instead. The joint fluid stays completely organism-free throughout. Every bit of the inflammation here is autoimmune, collateral damage in a fight that was actually won somewhere else entirely.



HLA-B27 presenting a self-antigen | T-cell attacking joint tissue directly

Still there: disseminated gonococcal infection

Disseminated gonococcal infection runs the opposite path entirely. *Neisseria gonorrhoeae* breaches the mucosal barrier and gets directly into the bloodstream. Live bacteria, not primed T-cells, do the traveling. It develops in only 0.5 to 3.0 percent of untreated gonococcal infections, but when it seeds a joint, it seeds it with the actual organism, and neutrophils flood in to fight a real, live infection.



Fully purulent joint fluid

The tap is what actually separates them

Septic arthritis must be actively ruled out before reactive arthritis is diagnosed. That means a tap happens first, every time, even when the story sounds textbook for reactive arthritis. Gonococcal arthritis is the most common cause of septic arthritis in otherwise healthy young adults in the United States, and it's exactly the population that also gets reactive arthritis. The rule that holds regardless: assume septic until proven otherwise. Treating a sterile joint like it's infected costs a few unnecessary antibiotic days. Treating an infected joint like it's sterile costs cartilage, and sometimes the joint itself.

Have something specific to your case?

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